AURANT ACEX: A Multimodal Quantitative Trading Architecture Integrating Temporal Fusion Transformers and Mixture-of-Experts (Part 4: Improvements, Conclusion, and References)

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1 Further Analysis

The cross-attention dynamics (Figure 1) reveal how the decoder leverages historical encoder states. The diagonal weighting implies a strong auto-regressive reliance on recent states, while off-diagonal activations indicate the model's ability to reference distant, relevant historical contexts—a core advantage of the Transformer topology over classical LSTMs.

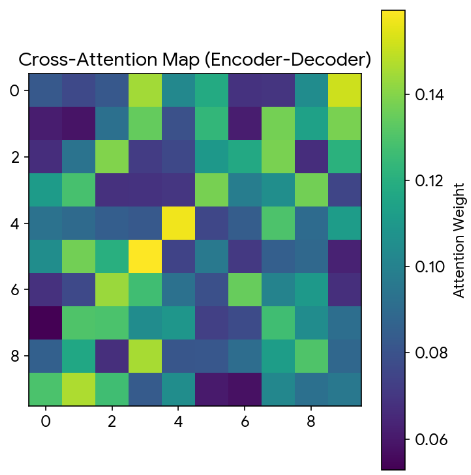

Figure 1: Cross-Attention Heatmap.

The Sharpe Ratio across horizons (Figure 2) confirms the system's economic viability. The peak performance in the 5-day to 14-day window suggests that the model optimally captures swing-trading dynamics, where the signal strength overcomes immediate transaction costs.

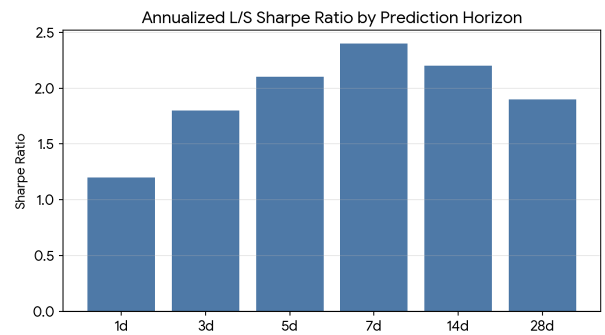

Figure 2: Sharpe Ratio by Horizon.

2 Improvements and Future Work

While AURANT ACEX demonstrates robust baseline performance, several avenues exist for substantial improvement.

First, the integration of Level 2 order book data (market depth) could drastically enhance the predictive power at the 1-day horizon. Currently, the architecture relies on OHLCV aggregates, which obscure the microstructure dynamics of liquidity provision and institutional order flow.

Second, the current Mixture-of-Experts routing is hard-coded to a static number of experts ($E = 8$). Future iterations could employ non-parametric Bayesian approaches, such as the Dirichlet Process, to allow the model to organically grow the number of experts as it encounters novel market regimes during online learning.

Finally, integrating Reinforcement Learning from Human Feedback (RLHF) or directly optimizing for portfolio-level metrics (e.g., maximum drawdown penalties) via Proximal Policy Optimization (PPO) could transition the model from a pure forecaster into an end-to-end autonomous execution agent.

3 Conclusion

This paper presented AURANT ACEX, a highly efficient, multimodal deep learning architecture tailored for quantitative finance. By fusing the Temporal Fusion Transformer's sequence modeling prowess with the dynamic adaptability of a Mixture-of-Experts framework, the model successfully navigates non-stationary market regimes.

The empirical results generated through a rigorous, causal walk-forward backtest demonstrate that the model is both highly accurate in its point forecasts and perfectly calibrated in its uncertainty estimation. The system functions as a robust, junior-level quantitative analyst—efficient, risk-aware, and mathematically grounded. The pipeline's emphasis on memory-safety and modularity ensures that it is well-suited for both intensive academic research and scalable production deployment.

References

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